

Link Analysis

By

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Structure of the web

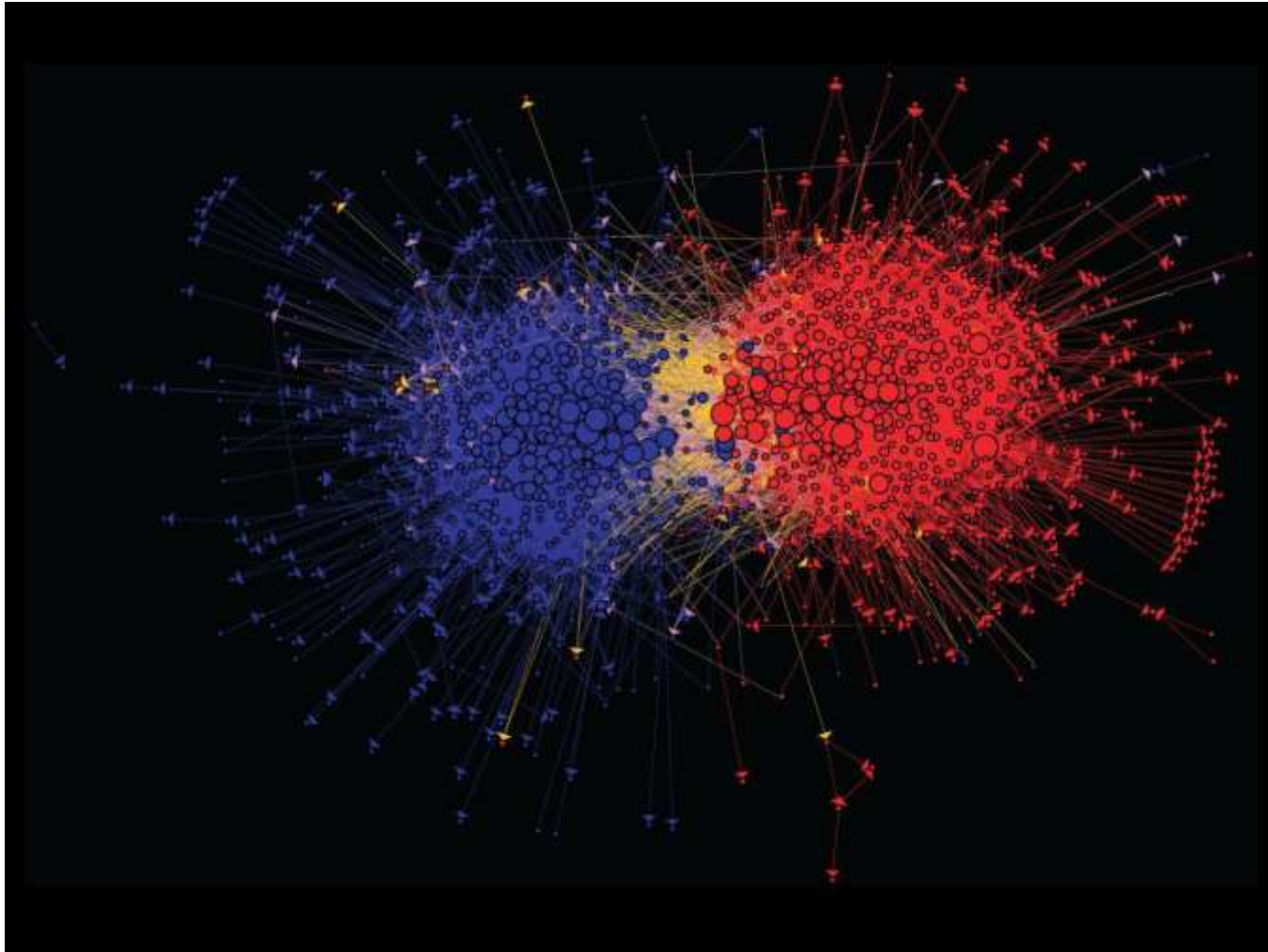
Graph Data: Social Networks



Facebook social graph

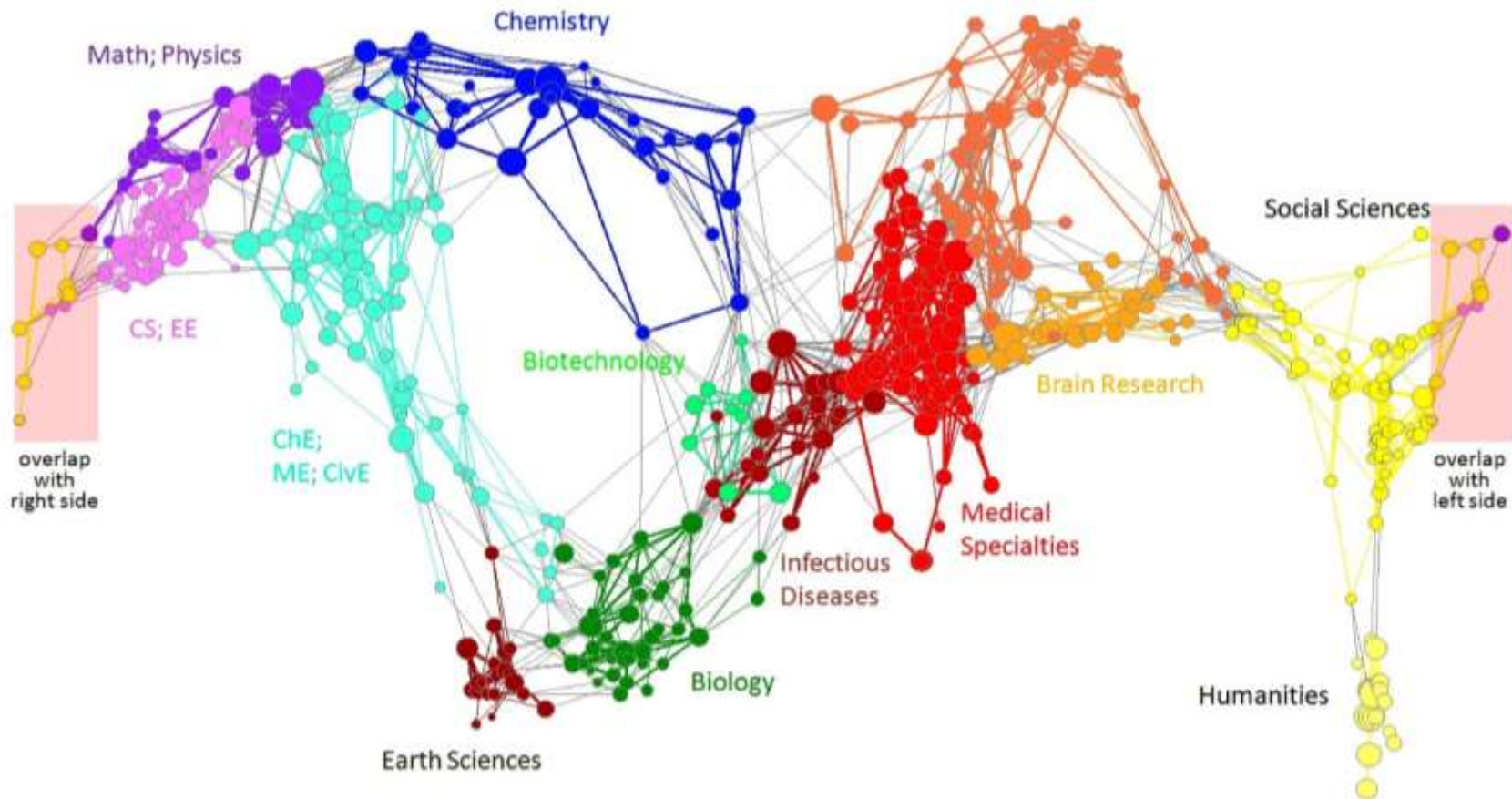
4-degrees of separation [Backstrom-Boldi-Rosa-Ugander-Vigna, 2011]

Graph Data: Media Networks



Connections between political blogs
J. Leskovec, A. Rajaraman, J. Ullman:
Polarization of the network [Adamic-Glance, 2005]
Mining of Massive Datasets,
<http://www.mmds.org>

Graph Data: Information Nets

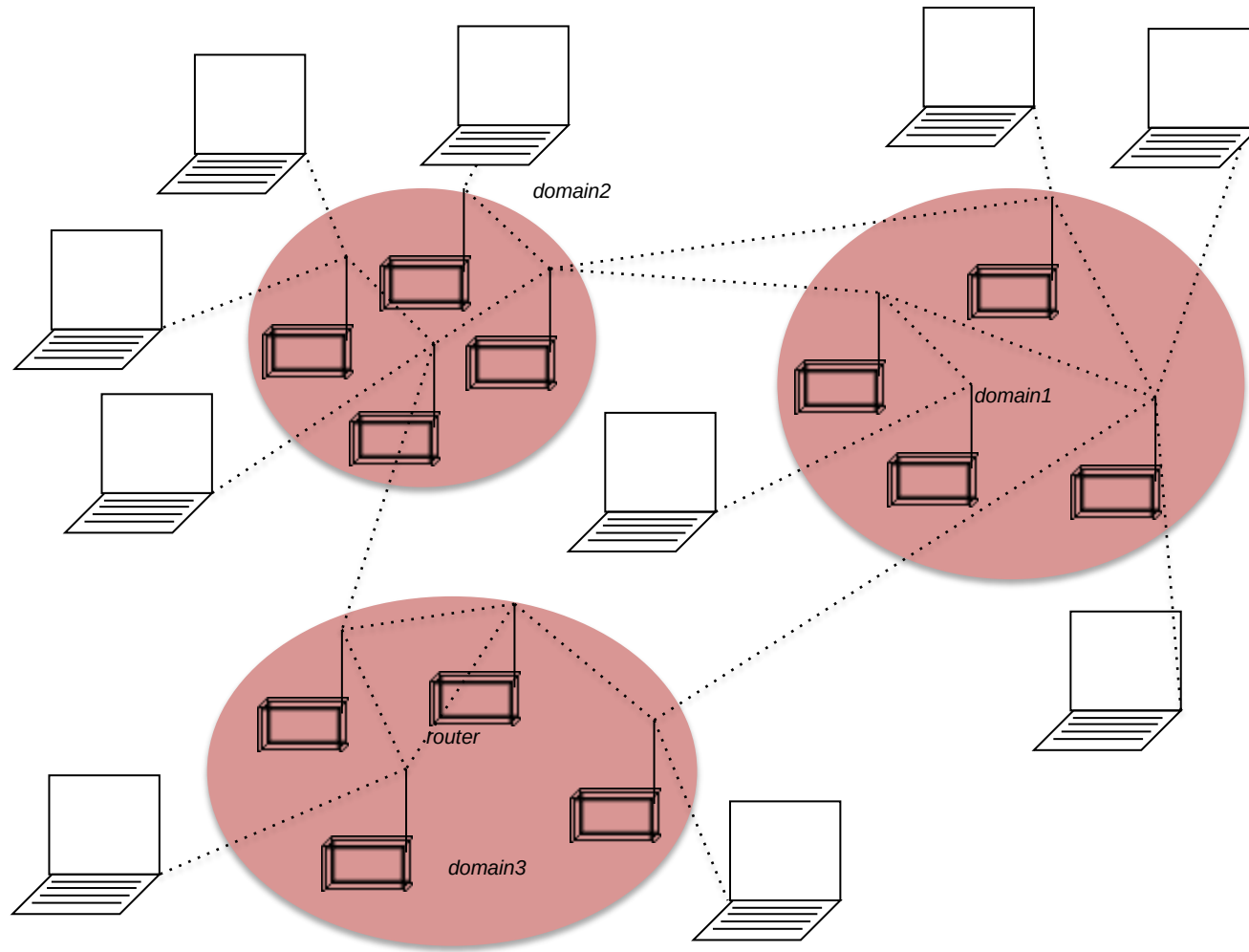


Citation networks and Maps of science

J. Leskovec, A. Rajaraman, J. Ullman:
[Börner et al., 2012]
Mining of Massive Datasets,

<http://www.mmds.org>

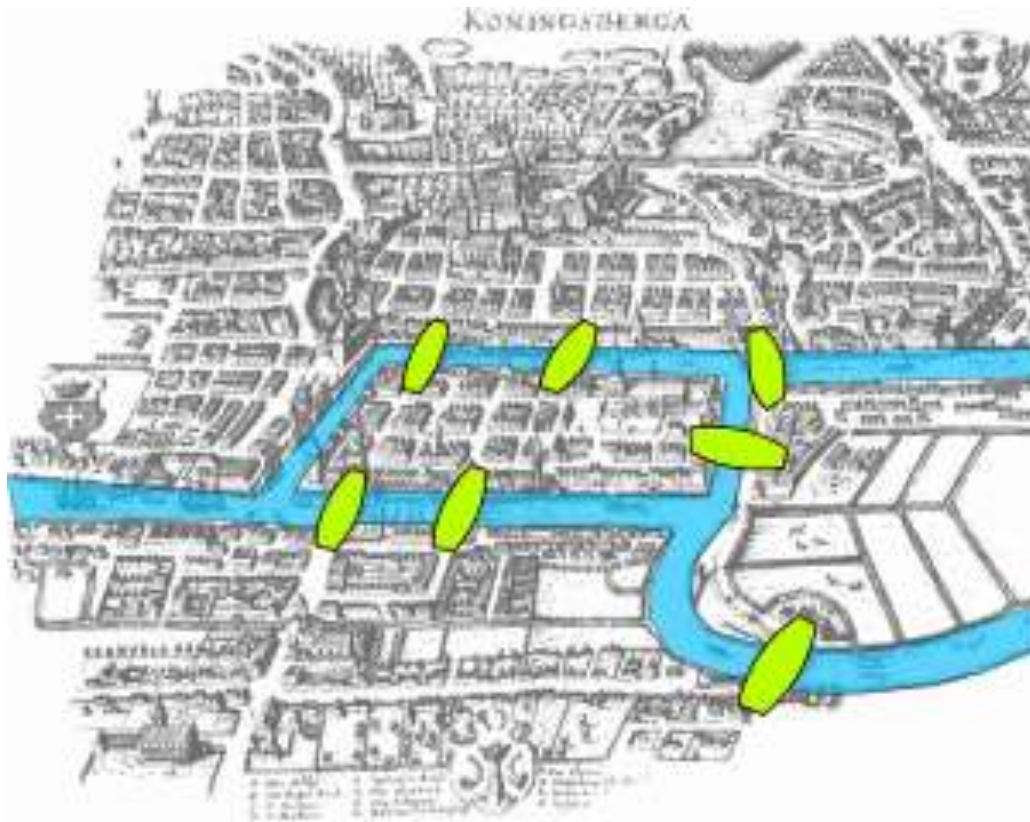
Graph Data: Communication Nets



Internet

J. Leskovec, A. Rajaraman, J. Ullman:
Mining of Massive Datasets,
<http://www.mmds.org>

Graph Data: Technological Networks



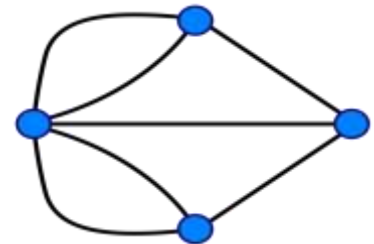
Seven Bridges of Königsberg [Euler, 1735]

Return to the starting point by traveling each link of the graph once and only once.

J. Leskovec, A. Rajaraman, J. Ullman:

Mining of Massive Datasets,

<http://www.mmds.org>



Web as a Graph

- **Web as a directed graph:**
 - **Nodes: Webpages**
 - **Edges: Hyperlinks**

I teach a
class on
Data
Structure.

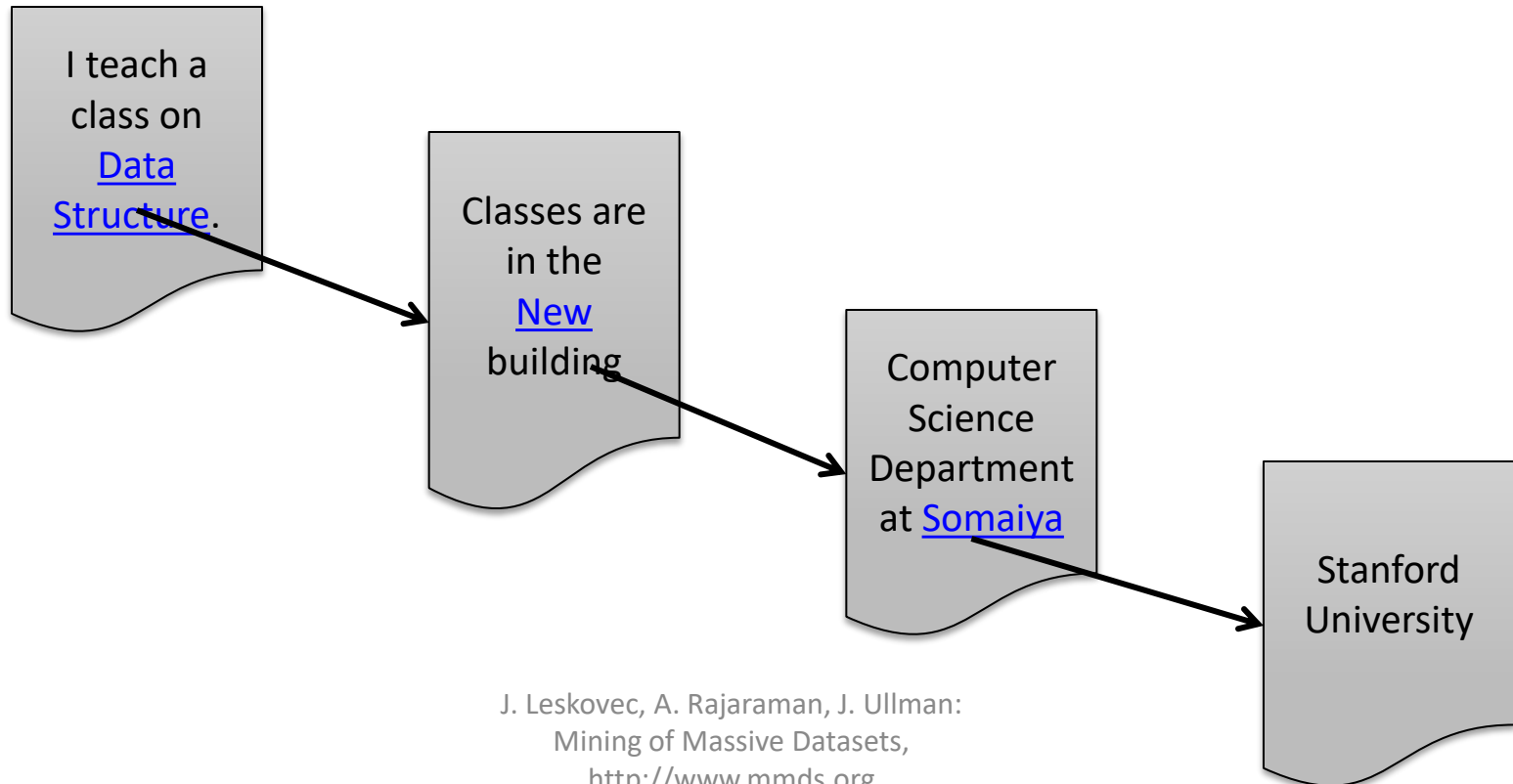
Classes are
in the
“New”
building

Computer
Engineering
Department

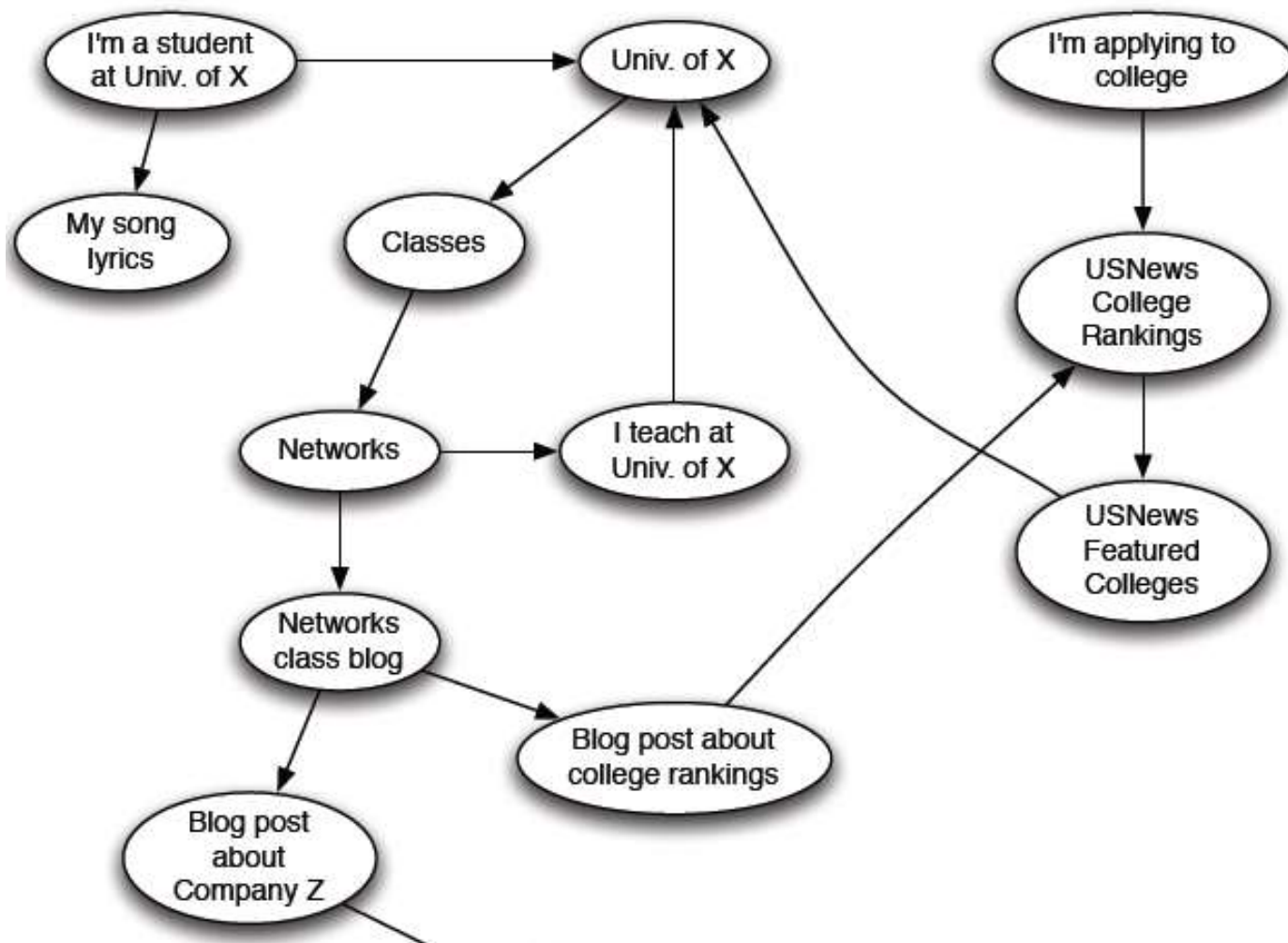
Somaiya
University

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Web as a Directed Graph



Broad Question

- **How to organize the Web?**
- **First try: Human curated Web directories**
 - Yahoo, DMOZ, LookSmart
- **Second try: Web Search**
 - **Information Retrieval** investigates:
Find relevant docs in a small and trusted set
 - Newspaper articles, Patents, etc.
 - **But:** Web is **huge**, full of untrusted documents, random things, web spam, etc.



Web Search: 2 Challenges

2 challenges of web search:

- (1) Web contains many sources of information
Who to “trust”?
 - **Trick:** Trustworthy pages may point to each other!
- (2) What is the “best” answer to query
“newspaper”?
 - No single right answer
 - **Trick:** Pages that actually know about newspapers might all be pointing to many newspapers

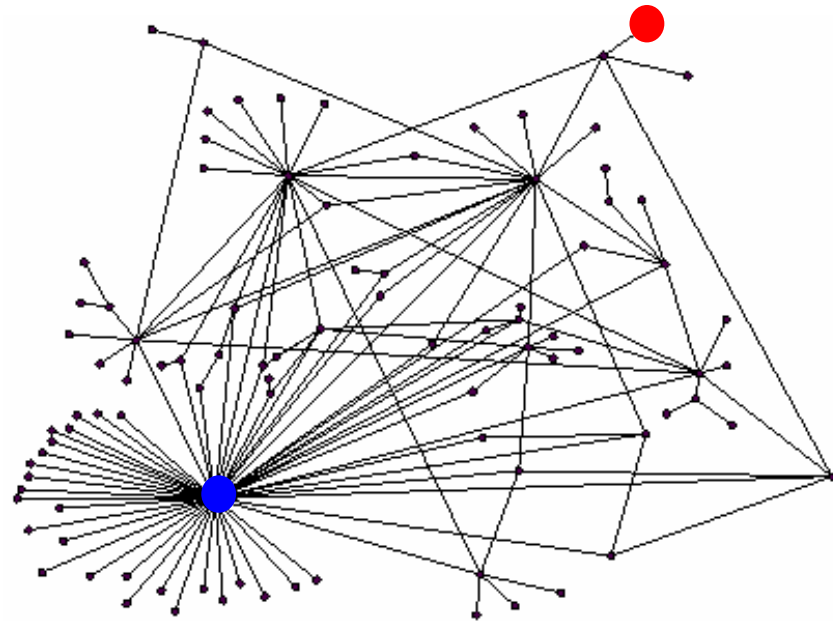
Ranking Nodes on the Graph

- All web pages are not equally “important”

[www.news-xyz.com](#) vs. [www.xyz.edu](#)

- There is large diversity in the web-graph node connectivity.

Let's rank the pages by the link structure!



Thank you